

When Dollar Strength Stops Widening the Basis: Regime-Dependent Hedging in the JPY–USD Swap Market

1. Introduction

For decades, the U.S. dollar has remained at the center of the international financial system as the dominant invoicing currency, the primary reserve asset, and the backbone of global funding and hedging markets. Because of this central role, fluctuations in the dollar transmit financial conditions across borders. A large and influential literature shows that dollar appreciations coincide with tighter global financial conditions, higher funding premia, and larger deviations from covered interest parity (CIP) (Bruno and Shin, 2015; Du, Tepper, and Verdelhan, 2018; Avdjiev, Du, Koch, and Shin, 2019). One of the most robust empirical regularities to emerge from this work is that a stronger dollar is associated with a wider cross-currency basis: when the dollar appreciates, the cost of obtaining dollars through FX swaps rises relative to direct dollar borrowing. Recent evidence, however, suggests that this relationship is more nuanced than previously understood. Rather than exhibiting a stable linear pattern, the sensitivity of the cross-currency basis to dollar movements appears to vary across states of the global financial cycle. ? provide early evidence that the response of basis spreads to dollar movements is regime dependent, showing that the effect of dollar appreciations weakens—and can even reverse—when the dollar is already in a strong regime. These findings challenge the conventional view that dollar strength uniformly tightens global funding conditions through the FX swap market and point instead to state-dependent transmission mechanism.

This paper builds on that insight by showing that understanding the regime dependence of the dollar–basis relationship requires focusing on a specific and economically central market: the JPY–USD cross-currency basis. Japan is the world’s largest net international creditor and home to some of the largest institutional investors in global bond markets. Japanese life insurers, pension funds, and banks hold substantial portfolios of U.S. dollar–denominated assets and actively manage the associated currency risk using FX swaps and forwards. As a result, the JPY–USD basis is not merely a measure of funding conditions, but the equilibrium price of large-scale currency hedging by Japanese institutions.

We document that the response of the JPY–USD basis to dollar shocks is strongly state dependent. The same dollar appreciation can have sharply different, and sometimes opposite, effects on the basis depending on the prevailing state of the dollar.

The puzzle. If dollar strength primarily reflects tighter global dollar liquidity and increased demand for dollar funding, then further dollar appreciations should uniformly widen the JPY basis. Yet, in the data, this is not what we observe. During some periods of pronounced dollar strength, additional dollar appreciations are associated with little change in the JPY basis—or

even with basis tightening. In contrast, during weaker dollar environments, appreciations are much more consistently associated with basis widening.

This instability is not a small-sample artifact. Using data on the JPY–USD cross-currency basis across maturities from 1 to 30 years, we show that the correlation between changes in the broad dollar and changes in the JPY basis frequently changes sign over time. Moreover, these sign switches are not random: they cluster in periods when the dollar is at historically high levels. The conventional linear view—“dollar up, basis widens”—fits the data well only in low-dollar environments, but breaks down when the dollar is already strong.

Main empirical finding. Our central empirical result is that the sensitivity of the JPY–USD cross-currency basis to dollar movements is regime dependent. In low-dollar regimes, a dollar appreciation leads to a statistically and economically significant widening of the JPY basis. In high-dollar regimes, this relationship weakens substantially and, at several maturities, reverses: further dollar strength is associated with basis tightening rather than widening.

This regime dependence is strong across maturities, precisely as Japanese life insurers and pension funds concentrate their FX-hedged U.S. bond exposures. The effect is weaker at the very long maturities, which are less liquid and more influenced by local supply–demand imbalances. These behavioral patterns across maturities point directly to institutional hedging flows rather than purely to short-term funding stress alone.

Why existing explanations fall short. The dominant interpretation of CIP deviations emphasizes limits to arbitrage arising from balance sheet costs, regulatory constraints, and funding liquidity risk (Du, Tepper, and Verdelhan, 2018; Sushko et al., 2016; ?). In these frameworks, a stronger dollar tightens intermediary constraints and raises the marginal cost of providing dollar funding, leading to a larger basis wedge. While this mechanism explains why the basis is typically negative and why it widens during episodes of acute stress, it has difficulty explaining why the marginal effect of dollar appreciation should weaken—or reverse—precisely when the dollar is already strong.

Funding-based explanations imply a roughly monotonic relationship between dollar strength and CIP deviations: if the dollar becomes more expensive, the basis should widen further. The JPY evidence instead reveals pronounced nonlinearity. Once the dollar reaches sufficiently high levels, additional appreciation does not generate the same widening pressure and may even coincide with basis compression.

Mechanism: regime-dependent hedging demand by Japanese institutions. We propose a complementary mechanism based on asset-side currency hedging by Japanese investors. In our framework, the JPY–USD basis reflects not only dealer balance sheet constraints but also equilibrium conditions in the market for currency hedges between Japanese holders of USD assets and global counterparties.

When the dollar is weak (a low-dollar regime), Japanese investors face heightened balance sheet sensitivity to further dollar depreciation and therefore maintain or increase FX hedging of their USD portfolios to limit currency risk. They therefore maintain or increase FX hedging of their large USD bond portfolios. An unexpected dollar appreciation in this regime raises the yen value of unhedged USD assets, mechanically increasing FX exposure relative to risk limits. As a result, institutions rebalance by raising hedge ratios—driven by risk management rules, solvency constraints, and asset–liability management—thereby increasing demand for dollars in swap markets and widening the basis.

When the dollar is already strong (a high-dollar regime), expectations shift. Japanese investors are more willing to remain partially unhedged, anticipating continued dollar strength

or higher FX carry. As a result, additional dollar appreciations generate much smaller increases in hedging demand and may even reduce net demand for dollars in swap markets, leading to a muted or reversed basis response. In this sense, the dollar has a “double life” in JPY markets: the same appreciation reflects heightened hedging demand in one regime and portfolio rebalancing in another.

Identification strategy. To test these predictions, we estimate threshold regressions that allow the effect of dollar movements on the JPY basis to vary across dollar regimes defined by the level of the broad dollar index (Hansen, 2000). We then estimate regime-conditional local projections to trace the dynamic response of the JPY basis to dollar shocks. In low-dollar regimes, dollar appreciations lead to persistent basis widening over subsequent weeks, consistent with gradual adjustments in institutional hedge ratios. In high-dollar regimes, the response is small and short-lived. We further show that regime asymmetry is strong across maturities associated with Japanese institutional hedging and is amplified when proxies for intermediary balance sheet constraints tighten.

Contributions. This paper makes three contributions.

First, we document a new stylized fact: the response of the JPY–USD cross-currency basis to dollar shocks is strongly regime dependent. The well-known “dollar appreciates, basis widens” result holds primarily in low-dollar environments and weakens or reverses when the dollar is already strong.

Second, we provide a theoretical framework linking these nonlinearities to regime-dependent hedging demand by Japanese institutional investors. By focusing on asset-side portfolio hedging, we interpret the JPY basis as a barometer of large-scale institutional currency-hedging flows.

Third, our findings have important implications for how policymakers and market participants interpret CIP deviations. In JPY markets, basis widening is not a uniform signal of dollar funding stress; its informational content depends on the prevailing dollar regime and the hedging behavior of Japanese institutions. Recognizing this asymmetry is essential for assessing global liquidity conditions and the international transmission of dollar shocks.

The remainder of the paper presents a brief review of relevant literature, develops the model, describes the data and empirical strategy, presents the main results, and discusses implications for international funding markets.

2. Literature Review

This paper contributes to four related strands of the literature: (i) the modern literature on deviations from covered interest parity (CIP) and cross-currency basis spreads, (ii) intermediary constraints and limits-to-arbitrage in dollar funding and FX swap markets, (iii) institutional investor currency hedging and portfolio rebalancing—particularly in Japan, and (iv) empirical approaches that allow for nonlinear, state-dependent transmission of shocks, including threshold methods and local projections.

2.1. CIP Deviations and the Cross-Currency Basis

A large post-crisis literature documents persistent and economically meaningful departures from CIP in FX swap markets, typically measured by the cross-currency basis (Du, Tepper, and Verdelhan, 2018; Avdjiev, Du, Koch, and Shin, 2019; ?). These deviations are especially pronounced for the U.S. dollar, consistent with its dominant role in global intermediation

and the international monetary system (Bruno and Shin, 2015). Early post-crisis accounts emphasized that CIP breakdowns reflect a wedge between synthetic dollar funding via swaps and direct dollar borrowing, and that this wedge can be large, persistent, and maturity-dependent (??).

While the empirical fact of CIP deviations is well established, the interpretation of basis movements remains an active area of research. A dominant view is that basis spreads reflect the price of balance sheet usage required to intermediate cross-currency positions, implying that basis widening is linked to stresses in dollar funding markets and tighter constraints on arbitrage capital (Du, Tepper, and Verdelhan, 2018; Avdjiev, Du, Koch, and Shin, 2019). We build on this literature but show that, even within the economically central JPY–USD market, the canonical “dollar appreciates, basis widens” relationship is not stable. Instead, the response of the basis to dollar shocks depends on the prevailing dollar regime.

2.2. Intermediary Constraints, Balance Sheet Costs, and Limits to Arbitrage

A second strand explains CIP deviations through intermediary constraints and limits to arbitrage. In this view, dealers and global banks absorb cross-currency order flow and provide synthetic dollar funding subject to balance sheet costs, regulatory constraints, and funding liquidity considerations (Du, Tepper, and Verdelhan, 2018). More broadly, intermediary asset-pricing frameworks emphasize that constrained financial intermediaries play a central role in pricing risk across markets (???). These models highlight how balance sheet tightness can amplify the price impact of order flow and generate time-varying wedges in otherwise arbitrage-linked markets.

Empirically, measures of intermediary tightness and funding stress—such as the LIBOR–OIS spread prior to post LIBOR, repo conditions, or dealer leverage capacity—are often associated with wider basis spreads, especially in episodes of market stress (Avdjiev, Du, Koch, and Shin, 2019). This perspective has generated major progress in understanding why CIP deviations persist even in deep markets. However, a pure funding-stress interpretation has difficulty rationalizing pronounced sign changes in the marginal response of basis spreads to dollar appreciation once the dollar is already strong. Funding-based models typically imply that stronger dollar states should be associated with higher marginal scarcity of dollars and therefore a monotone negative relationship between dollar strength and basis widening on average. Our evidence for Japan instead reveals that the slope can attenuate sharply and even reverse across dollar regimes, motivating our focus on a complementary mechanism: state-dependent asset-side hedging demand, with intermediary constraints acting primarily as an amplification layer.

2.3. Institutional Investor Hedging and Asset-Side Demand for FX Swaps

A growing literature emphasizes that the cross-currency basis is shaped not only by dealer constraints but also by the hedging needs of global investors holding large cross-border bond portfolios (Du, Tepper, and Verdelhan, 2018; ?). The currency composition of international portfolios and the cost of hedging materially affect realized returns and capital allocation decisions (?). At the firm level, currency exposures and hedging choices are also shown to be state dependent and linked to balance sheet considerations (?).

These asset-side forces are particularly important for Japan. Japanese life insurers, pension funds, and banks, including its central bank, are among the largest global holders of U.S. dollar-denominated bonds and actively manage FX risk using swaps and forwards. Institutional features—such as regulatory capital rules, accounting treatment, and asset–liability

management constraints—shape the speed and intensity of hedge adjustments, often generating gradual and state-dependent changes in hedging demand. Evidence from Japanese institutional portfolios and hedging behavior supports the view that large buy-side flows can exert sustained pressure on FX swap markets, especially at medium and long maturities (?). These features align closely with our dynamic findings: in low-dollar regimes, basis widening following dollar shocks builds over several weeks, consistent with gradual hedge-ratio adjustment rather than purely contemporaneous funding stress.

2.4. Japan, the JPY–USD Basis, and the Global Dollar Cycle

Japan plays a unique role in global funding markets as the world’s largest net international creditor and a key supplier of cross-border capital. The global financial cycle literature shows that U.S. monetary conditions and dollar fluctuations transmit internationally through balance sheets, risk-taking, and capital flows (?Bruno and Shin, 2015; ?). In asset-pricing terms, global dollar risk is associated with time variation in expected returns and intermediary risk-bearing capacity (?). The JPY–USD basis is therefore a particularly informative object for understanding how global dollar shocks are transmitted through both funding and portfolio-hedging channels.

Our contribution is to show that the transmission of dollar shocks into the JPY–USD basis is nonlinear in a way that naturally aligns with institutional hedging incentives. When the dollar is weak or moderate, a positive dollar shock tends to induce stronger hedging demand for USD exposures, widening the basis persistently. When the dollar is already strong, hedging incentives shift—for example, due to expectations of continued dollar strength or a greater willingness to accept FX carry—and the marginal hedging response attenuates, so basis responses become muted or even reverse. This state dependence implies that basis widening is not necessarily a uniform proxy for marginal funding stress: its informational content depends on the dollar regime and the underlying configuration of hedging demand and swap supply.

2.5. Nonlinear and State-Dependent Empirical Methods

A final strand of literature concerns empirical methods designed to capture nonlinear, state-dependent effects. Threshold models provide a disciplined way to allow effects to differ across regimes determined by an observable variable, with formal inference procedures for threshold estimation (Hansen, 2000). In macro-finance applications, such regime dependence often reflects nonlinear behavior in risk premia, constraints, or expectations. We use threshold regressions to allow the effect of dollar shocks on the basis to differ in low- versus high-dollar states, and we show that the regime boundary is stable and economically interpretable.

To trace dynamic adjustments, we complement threshold regressions with regime-conditional local projections (?). Local projections are well suited for estimating impulse responses without imposing strong parametric restrictions and have become standard in settings where dynamics may differ across states. Our regime-conditional impulse responses provide direct evidence that the basis response is persistent in low-dollar regimes but short-lived in high-dollar regimes, consistent with gradual institutional hedging adjustment and with intermediary constraints scaling—rather than determining—the transmission of net hedging pressure into equilibrium basis movements.

2.6. Positioning and Contribution

Taken together, the existing literature provides two core insights: (i) CIP deviations and basis spreads reflect limits to arbitrage and balance sheet costs in global dollar intermediation, and (ii) hedging demand from large cross-border investors can be a first-order driver of swap-market pricing. This paper contributes by showing that, in Japan’s systemically important JPY–USD market, the mapping from dollar shocks to basis movements is not stable but strongly regime dependent. We interpret the sign reversal as evidence of state-dependent institutional hedging incentives, with intermediary tightness amplifying the equilibrium impact of net hedging order flow. Methodologically, we combine threshold inference (Hansen, 2000) with regime-conditional local projections (?) to quantify both contemporaneous and dynamic regime asymmetries across maturities most relevant for Japanese life insurers and pension funds and other hedged Japanese holders of US assets.

3. Stylized Facts: Dollar Regimes and the JPY Basis

This section documents three empirical regularities that motivate our regime-dependent hedging framework in the JPY–USD market. First, the JPY–USD cross-currency basis is tightly linked to movements in the broad U.S. dollar, but the relationship is far from stable. Second, the dollar–basis relationship exhibits pronounced regime dependence: the same dollar appreciation is associated with basis widening in some periods and basis tightening in others. Third, this nonlinearity is strong across maturities, as Japanese investors are active in currency hedging markets across the term structure.

3.1. The JPY Basis and the Dollar

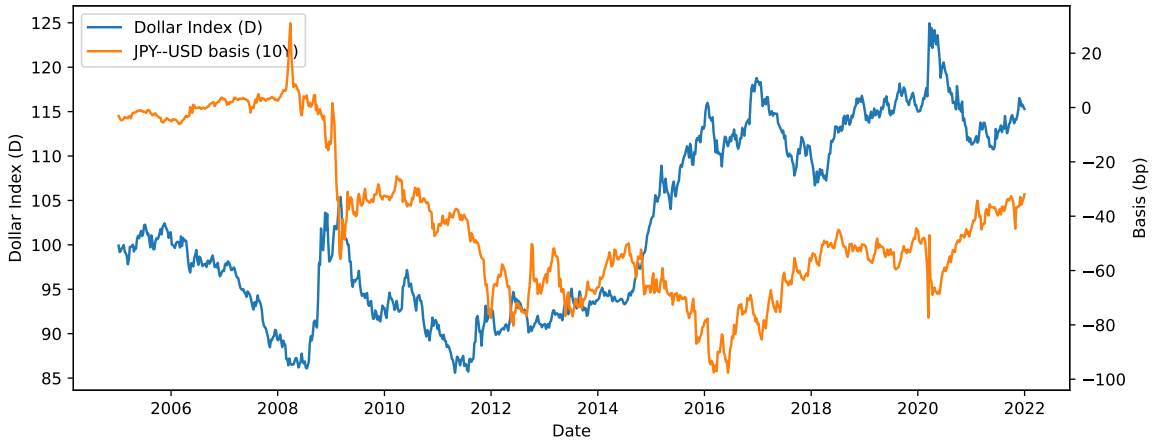


Figure 1: Broad dollar index and the 10-year JPY–USD cross-currency basis.

We begin by documenting the unconditional relationship between the broad dollar and the JPY–USD cross-currency basis. Figure 1 plots the broad dollar index alongside the 10-year JPY basis. While episodes of dollar strength often coincide with basis widening, there are extended periods in which dollar appreciations are associated with basis compression instead. The co-movement is therefore visible but clearly unstable.

To quantify this relationship, we compute unconditional correlations between weekly changes in the broad dollar index (ΔD_t) and changes in the JPY basis ($\Delta x_{JPY,n,t}$) across maturities n . Although the average correlation is negative—consistent with the conventional “dollar up, basis widens” narrative—the magnitude and even the sign of the relationship vary substantially across time and maturities. This dispersion indicates that a single linear slope cannot adequately characterize the dollar–basis linkage in JPY markets.

3.2. Summary Statistics Across Maturities

To provide basic context on the JPY–USD basis curve, Table 1 below reports summary statistics for the basis across maturities from 1 to 30 years, together with cross-maturity correlations of weekly changes.

Three patterns emerge. First, the JPY basis is negative on average across all maturities, consistent with persistent deviations from covered interest parity in dollar funding via swaps. Second, volatility is concentrated in the belly of the curve (5–10 years), while the very long end is comparatively smoother. Third, cross-maturity comovement is strongly maturity-segmented: the correlation of weekly changes is highest among long maturities (10–30 years) but substantially lower between the short end (1 year) and the long end. This mirrors the key stylized fact emphasized in related work on the EUR/USD basis: correlations between the short and long end are much lower than correlations within the long end. (Ibhagui, 2021)

Table 1: JPY–USD basis summary statistics and cross-maturity correlations (full sample)

	1Y	5Y	10Y	20Y	30Y
Mean	-26.52	-42.32	-42.69	-32.66	-27.02
Median	-26.25	-45.80	-50.00	-37.00	-22.60
Std. Dev.	18.49	27.95	27.89	21.43	20.24
Range	85.62	120.50	128.62	108.00	102.75
Minimum	-79.62	-101.50	-97.62	-75.00	-74.75
Maximum	6.00	19.00	31.00	33.00	28.00
<i>Correlation matrix of weekly changes</i>					
	1Y	5Y	10Y	20Y	30Y
1Y	1.00	0.63	0.42	0.33	0.33
5Y	0.63	1.00	0.86	0.71	0.65
10Y	0.42	0.86	1.00	0.88	0.81
20Y	0.33	0.71	0.88	1.00	0.95
30Y	0.33	0.65	0.81	0.95	1.00

Notes: Summary statistics are in basis points for the JPY–USD cross-currency basis at each maturity. The correlation matrix reports correlations of weekly changes.

Table 2: JPY–USD basis summary statistics (Low-dollar regime, $D_t \leq 111.81$)

	1Y	5Y	10Y	20Y	30Y
Mean	-22.47	-38.20	-37.00	-25.04	-18.73
Median	-22.50	-45.00	-40.50	-25.00	-14.25
Std. Dev.	17.26	29.64	29.44	19.89	17.04
Range	80.00	118.80	128.62	104.50	90.90
Minimum	-74.00	-99.80	-97.62	-71.50	-62.90
Maximum	6.00	19.00	31.00	33.00	28.00
<i>Correlation matrix of weekly changes</i>					
	1Y	5Y	10Y	20Y	30Y
1Y	1.00	0.62	0.38	0.29	0.30
5Y	0.62	1.00	0.83	0.62	0.55
10Y	0.38	0.83	1.00	0.83	0.74
20Y	0.29	0.62	0.83	1.00	0.93
30Y	0.30	0.55	0.74	0.93	1.00

Notes: Summary statistics are in basis points for the JPY–USD cross-currency basis at each maturity. The correlation matrix reports correlations of weekly changes.

Table 3: JPY–USD basis summary statistics (High-dollar regime, $D_t > 111.81$)

	1Y	5Y	10Y	20Y	30Y
Mean	-37.18	-53.17	-57.71	-52.73	-48.86
Median	-30.94	-46.75	-53.34	-51.06	-47.56
Std. Dev.	17.38	19.07	15.30	8.43	8.10
Range	65.00	76.50	65.62	36.25	44.88
Minimum	-79.62	-101.50	-97.50	-75.00	-74.75
Maximum	-14.62	-25.00	-31.88	-38.75	-29.88
<i>Correlation matrix of weekly changes</i>					
	1Y	5Y	10Y	20Y	30Y
1Y	1.00	0.69	0.50	0.45	0.41
5Y	0.69	1.00	0.93	0.87	0.82
10Y	0.50	0.93	1.00	0.95	0.91
20Y	0.45	0.87	0.95	1.00	0.97
30Y	0.41	0.82	0.91	0.97	1.00

Notes: Summary statistics are in basis points for the JPY–USD cross-currency basis at each maturity. The correlation matrix reports correlations of weekly changes.

Tables 2 and 3 repeat the same diagnostics by dollar regime and show that both the level and cross-maturity comovement of the basis differ across states of the dollar.

3.3. Instability and Sign Switching

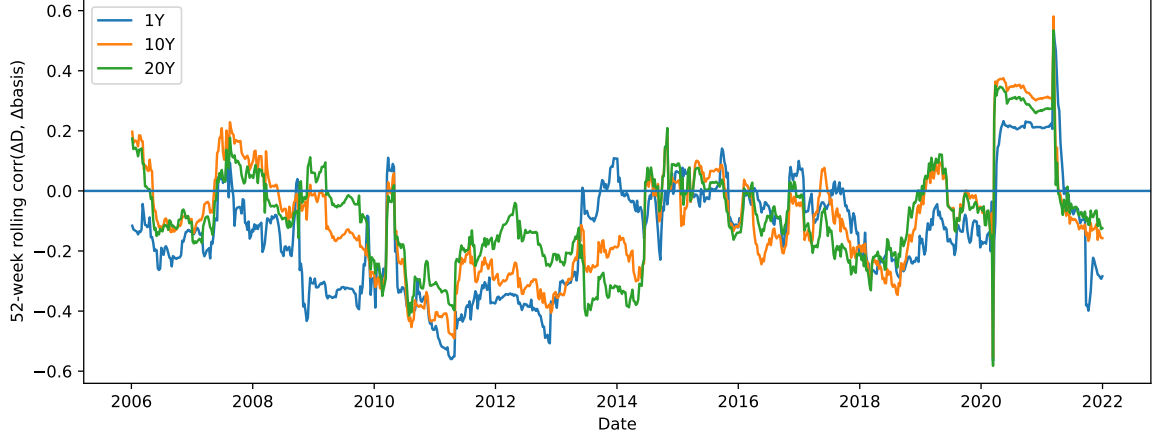


Figure 2: Rolling 52-week correlations between weekly changes in the broad dollar and weekly changes in the JPY basis (selected maturities).

The instability of the dollar–JPY basis relationship is evident in rolling correlation analysis. Figure 2 shows 52-week rolling correlations between ΔD_t and $\Delta x_{JPY,n,t}$ for 1-, 10-, and 20-year maturities.

Two features stand out. First, the correlation frequently changes sign. Over the full sample, the JPY basis exhibits more than a dozen sign switches in its rolling correlation with the dollar, a pattern that is broadly comparable across maturities. This level of directional instability is inconsistent with a stable linear relationship.

Second, sign reversals cluster in periods when the dollar is at historically elevated levels. During these high-dollar episodes, further dollar appreciations are often associated with JPY basis tightening rather than widening. In contrast, in lower-dollar environments, appreciations are more consistently linked to basis widening. These patterns suggest that the dollar–basis relationship is not merely noisy, but systematically state dependent.

3.4. Regime Dependence in the Dollar–JPY Basis Relationship

To formalize this state dependence, we estimate separate regressions of $\Delta x_{JPY,n,t}$ on ΔD_t in low- and high-dollar subsamples defined by the estimated dollar regime threshold.

The results reveal a striking asymmetry. In the low-dollar regime, dollar appreciations are associated with statistically significant basis widening across most maturities, with particularly strong effects across tenors. In the high-dollar regime, however, the estimated coefficients turn positive, indicating basis tightening rather than widening. This sign reversal is pronounced across maturities, where the magnitude of the high-regime coefficient is not only significant but often of opposite sign relative to the low-regime estimate.

These findings show that the canonical relationship—dollar appreciates, the basis widens—is not a structural constant in the JPY markets. Instead, the effect of dollar movements on the JPY basis depends critically on the prevailing dollar regime.

These results remain strong at long maturities. This segment of the curve corresponds closely to the hedging activity of major Japanese institutional investors, including life insurers

and pension funds, which hold substantial portfolios of foreign-currency bonds and actively hedge U.S. dollar exposures using medium- and long-dated FX swaps. Ultra-long maturities, however, are less liquid and more influenced by local supply–demand imbalances, so the estimated effects there are typically smaller. By contrast, shorter and intermediate maturities often exhibit stronger responses, consistent with portfolio managers’ preference for rolling hedges, which are naturally concentrated at shorter tenors.

Taken together, these stylized facts point to a clear conclusion: the relationship between the dollar and the JPY–USD cross-currency basis is inherently nonlinear, state dependent, and strongest at maturities where Japanese institutional buy-side hedging flows are most active. These empirical regularities directly motivate the regime-dependent hedging model developed in the next section.

4. Model and Testable Implications

4.1. Environment, Timing, and Dollar Regimes

Fix a currency pair $(f, \$)$ and a swap maturity n (suppressed for notation). Let s_t denote the log spot exchange rate (USD per unit of foreign currency), so that $\Delta s_{t+1} > 0$ corresponds to a USD depreciation (foreign currency appreciation). Let x_t denote the cross-currency basis at maturity n , using the convention that $x_t < 0$ represents a *negative basis* (a USD premium in FX swaps) and that a decline in x_t (more negative) corresponds to *basis widening*.

Time is weekly. The economy switches between two dollar regimes $\theta_t \in \{L, H\}$ determined by the level of a broad USD index D_t relative to a threshold D^τ :

$$\theta_t = \begin{cases} L, & D_t < D^\tau, \\ H, & D_t \geq D^\tau. \end{cases} \quad (1)$$

We interpret L as a low-/intermediate-dollar regime and H as a high-dollar regime.

A primitive state variable governing hedging incentives is the regime-conditional expectation of next-period exchange rate change,

$$\mu_t^\theta \equiv \mathbb{E}_t[\Delta s_{t+1} \mid \theta_t = \theta], \quad (2)$$

and $\sigma_{s,t}^2 \equiv \text{Var}_t(\Delta s_{t+1})$. We introduce a dollar-strength shock u_t (empirically proxied by ΔD_t) to capture current dollar movements, so that $u_t > 0$ corresponds to a broad dollar appreciation.

4.2. Agents and Hedging Motives

The mechanism is asset-side FX hedging. It features: (i) non-USD investors (i.e. Japanese holders of US assets) who hedge USD-denominated assets and therefore *demand USD* in FX swap markets, (ii) USD-based investors who hedge foreign-currency assets and therefore *supply USD* in FX swap markets, and (iii) constrained intermediaries who clear net USD swap order flow at an equilibrium basis.

Hedge positions are modeled as interior solutions to mean–variance problems. They represent hedge positions scaled by exposure.

Non-USD investors holding USD assets (USD-demanding hedgers)... A representative non-USD investor holds USD-denominated assets with USD notional $A_t^F > 0$ and chooses a hedge ratio h_t^F .

The optimal hedge component implied by mean–variance preferences is

$$h_t^{F,opt} = 1 + \frac{\mu_t^\theta - \pi_t}{\gamma_F \sigma_{s,t}^2}, \quad \gamma_F > 0, \quad (3)$$

where π_t is the effective marginal cost of hedging via swaps.

Institutional features such as risk limits, collateral constraints, benchmark tracking, and hedging mandates generate additional shock-driven hedge adjustments. We capture these with a reduced-form hedge shift rule:

$$h_t^F = h_t^{F,opt} + \eta^\theta u_t, \quad (4)$$

where η^θ define across dollar regimes. This reduced-form shift becomes:

$$h_t^F = 1 + \frac{\mu_t^\theta - \pi_t}{\gamma_F \sigma_{s,t}^2} + \eta^\theta u_t, \quad \gamma_F > 0, \quad (5)$$

where η^θ may differ across dollar regimes, where π_t is the effective marginal cost of hedging via swaps/forwards at maturity n (defined below). The term $\eta^\theta u_t$ captures shock-driven adjustments in hedge ratios arising from institutional features such as risk limits, benchmark constraints, margin dynamics, and hedging mandates. These mechanical rebalancing responses operate in addition to forward-looking portfolio considerations and may differ across dollar regimes.

Hedging USD assets corresponds to selling USD forward (buying home currency forward). We normalize flows so that higher h_t^F increases net USD intermediation demand in FX swap markets. The resulting USD notional demand for swap positions is

$$Q_t \equiv A_t^F h_t^F, \quad (6)$$

measured in USD.

USD-based investors holding foreign assets (USD-supplying hedgers)... A representative USD investor holds foreign-currency assets with USD-equivalent notional $A_t^U > 0$ and chooses a hedge ratio h_t^U as follows:

The optimal hedge component is

$$h_t^{U,opt} = 1 - \frac{\mu_t^\theta + \pi_t^U}{\gamma_U \sigma_{s,t}^2}, \quad \gamma_U > 0, \quad (7)$$

where π_t^U is the effective marginal cost/benefit of hedging foreign exposure.

Institutional rebalancing generates shock-driven adjustments:

$$h_t^U = h_t^{U,opt} + \xi^\theta u_t, \quad (8)$$

where ξ^θ may vary across regimes.

$$h_t^U = 1 - \frac{\mu_t^\theta + \pi_t^U}{\gamma_U \sigma_{s,t}^2} + \xi^\theta u_t, \quad \gamma_U > 0, \quad (9)$$

where π_t^U is the effective marginal cost/benefit of hedging foreign-currency exposure using swaps. The term $-\mu_t^\theta$ reflects that expected foreign-currency appreciation makes unhedged foreign exposure more attractive for a USD-based investor, reducing hedging demand. The term $\xi^\theta u_t$ allows for regime-dependent, shock-driven hedge adjustments due to institutional rebalancing.

Hedging foreign assets corresponds to buying USD forward (selling foreign currency forward). We normalize flows so that higher h_t^U increases USD supply to swap markets. The resulting USD notional supply of swap positions is

$$R_t \equiv A_t^U h_t^U, \quad (10)$$

also measured in USD.

4.3. Intermediary Clearing and the Basis as the Price of Net USD Order Flow

Net USD order flow in FX swaps is

$$N_t \equiv Q_t - R_t, \quad (11)$$

so that $N_t > 0$ represents net USD demand in swap markets. In this case, intermediaries must provide additional USD balance sheet capacity, and the basis becomes more negative (a wider negative basis).

Intermediaries absorb N_t subject to convex balance-sheet costs. We use a reduced-form clearing condition in which the basis compensates intermediaries for the marginal cost of accommodating net order flow:

$$x_t = -\Lambda N_t, \quad \Lambda > 0, \quad (12)$$

where Λ summarizes intermediary constraints (balance-sheet capacity, regulatory costs, and risk limits). The parameter Λ may vary over time and will be proxied in the empirics.

Finally, investors' effective marginal hedging costs depend on the basis:

$$\pi_t = \pi_0 - \kappa x_t, \quad \pi_t^U = \pi_0^U + \kappa_U x_t, \quad \kappa, \kappa_U > 0. \quad (13)$$

A more negative basis (lower x_t) raises hedging costs for non-USD investors (higher π_t) but increases the compensation to USD-based investors supplying USD through swaps (lower effective hedging cost through π_t^U).

4.4. Closed-Form Equilibrium

Substituting (5)–(9) and (13) into (11) and imposing market clearing (12) yields a unique linear equilibrium for the basis.

Define

$$\Psi_t \equiv \left(\frac{A_t^F \kappa}{\gamma_F \sigma_{s,t}^2} + \frac{A_t^U \kappa_U}{\gamma_U \sigma_{s,t}^2} \right) > 0, \quad \Xi_t^\theta \equiv A_t^F \eta^\theta - A_t^U \xi^\theta, \quad (14)$$

and

$$\Omega_t^\theta \equiv \left(\frac{A_t^F}{\gamma_F \sigma_{s,t}^2} + \frac{A_t^U}{\gamma_U \sigma_{s,t}^2} \right) \mu_t^\theta + \left(\frac{-A_t^F \pi_0}{\gamma_F \sigma_{s,t}^2} + \frac{A_t^U \pi_0}{\gamma_U \sigma_{s,t}^2} \right) + (A_t^F - A_t^U). \quad (15)$$

Proposition 1 (Equilibrium basis). *Suppose $\Lambda > 0$ and $\Psi_t \geq 0$. Then the equilibrium exists and is unique, with basis*

$$x_t = -\frac{\Lambda}{1 + \Lambda \Psi_t} \Omega_t^\theta - \frac{\Lambda}{1 + \Lambda \Psi_t} \Xi_t^\theta u_t. \quad (16)$$

Economic implication (Proposition 1). Equation (16) decomposes basis movements into two components. The first term captures how regime-dependent FX expectations μ_t^θ and baseline hedging wedges map into equilibrium net USD hedging pressure. The second term isolates the *dollar-shock transmission channel*: a dollar appreciation affects the basis through regime-dependent net hedging intensity Ξ_t^θ scaled by the intermediary multiplier $\Lambda/(1 + \Lambda \Psi_t)$. This delivers a direct mechanism-to-data bridge: state dependence in the dollar–basis relationship is equivalent to state dependence in net hedging intensity.

Proposition 2 (Regime-dependent dollar–basis slope and sign flips). *Let $\beta_t^\theta \equiv \partial x_t / \partial u_t \mid \theta_t = \theta$. Then*

$$\beta_t^\theta = -\frac{\Lambda}{1 + \Lambda \Psi_t} \Xi_t^\theta = -\frac{\Lambda}{1 + \Lambda \Psi_t} (A_t^F \eta^\theta - A_t^U \xi^\theta). \quad (17)$$

Hence:

1. If $A_t^F \eta^\theta > A_t^U \xi^\theta$, then $\beta_t^\theta < 0$: a dollar appreciation widens the (negative) basis.
2. If $A_t^F \eta^\theta < A_t^U \xi^\theta$, then $\beta_t^\theta > 0$: a dollar appreciation tightens the basis.
3. If $\eta^L > \eta^H$ and $\xi^H \geq \xi^L$, then $|\beta_t^L| > |\beta_t^H|$, delivering attenuation and potentially a sign flip in the high-dollar regime.

Economic implication (Proposition 2). The sign of the dollar–basis slope is pinned down by which side dominates *net* USD hedging pressure. In low-dollar regimes, a positive dollar shock can increase the hedging intensity of non-USD holders of USD assets (large η^L), raising net USD demand N_t and making the basis more negative. In high-dollar regimes, if non-USD investors relax hedges (smaller η^H) and/or USD investors increase hedging of foreign exposure (larger ξ^H), net USD demand can fall and the basis response attenuates or reverses. This mechanism is inherently asset-side: it links the basis to equilibrium currency-hedging flows rather than to borrower-side USD funding needs alone.

Proposition 3 (Intermediary constraints amplify regime effects). *For $\Psi_t \geq 0$, the magnitude of the dollar–basis response is increasing in intermediary constraints Λ :*

$$|\beta_t^\theta| = \frac{\Lambda}{1 + \Lambda \Psi_t} |A_t^F \eta^\theta - A_t^U \xi^\theta| \quad \text{is increasing in } \Lambda. \quad (18)$$

Economic implication (Proposition 3). Intermediary constraints are an *amplification layer*: they scale the translation of hedging-driven net order flow into the equilibrium basis. When balance-sheet capacity tightens (higher Λ), the same hedging imbalance generates larger basis movements in both regimes. This implies that the regime-dependent slope reversal can coexist with, and is strengthened by, periods of intermediary stress, but cannot be reduced to a funding-stress story alone because the sign of the response is governed by net hedging intensity.

4.5. Testable Implications

The model yields three empirical restrictions that map directly to our threshold and interaction regressions.

(i) *State-dependent slopes..* With u_t proxied by ΔD_t , Proposition 2 implies

$$\beta_{i,n}^L < 0, \quad \beta_{i,n}^H \text{ is closer to zero or positive,} \quad \beta_{i,n}^H - \beta_{i,n}^L > 0, \quad (19)$$

consistent with attenuation and potential sign flips at high dollar levels.

(ii) *Tenor pattern consistent with institutional hedgers..* Regime asymmetry should be strongest at tenors where long-horizon institutional hedgers dominate swap demand and supply, i.e., where variation in $A_t^F \eta^\theta$ is economically important. This motivates testing whether

$$\left| \beta_{i,n}^L \right| - \left| \beta_{i,n}^H \right|$$

is largest at intermediate maturities (e.g., 5–10y) relative to the front end and ultra-long end.

(iii) *Amplification by intermediary tightness..* Proposition 3 motivates interacting dollar shocks with an intermediary-tightness proxy Λ_t and testing whether the interaction term is economically meaningful and shares the sign of the baseline dollar coefficient within each regime.

5. Data and Empirical Strategy

5.1. Data and Variable Construction

Cross-currency basis.. Our primary dependent variable is the JPY–USD cross-currency basis swap spread at maturity n , denoted $x_{n,t}$. We adopt the market convention that a *more negative* value corresponds to a *wider* basis (a higher premium to obtain USD via swaps). Weekly changes are defined as $\Delta x_{n,t} = x_{n,t} - x_{n,t-1}$.

We examine maturities $n \in \{1, 2, 5, 10, 20, 30\}$ years to capture short-, medium-, and long-term hedging pressures relevant for Japanese institutional investors.

Dollar movements.. Dollar strength is measured using the Broad U.S. Dollar Index, D_t . Dollar shocks are defined as weekly changes $\Delta D_t = D_t - D_{t-1}$, where $\Delta D_t > 0$ denotes dollar appreciation.

Dollar regimes.. Following the framework developed in Section ??, regimes are determined by the *contemporaneous level* of the broad dollar index relative to an estimated threshold $\mathcal{D}_\tau$:

$$\text{Low-dollar regime: } D_t < \mathcal{D}_\tau, \quad \text{High-dollar regime: } D_t \geq \mathcal{D}_\tau.$$

This definition reflects the economic interpretation of regimes as states of the global dollar cycle rather than predetermined conditions. A dollar appreciation occurring when the dollar is already strong may have different implications than the same appreciation when the dollar is weak.

Control variables.. We include standard controls used in the CIP deviation literature:

$$Z_t = [\Delta \text{VIX}_t, \Delta \text{IV}_t, \Delta \text{RR}_t, \Delta \text{YSPRD}_t, \Delta \text{TSPRD}_t]'$$

These variables capture global risk sentiment, FX market conditions, and interest rate differentials between Japan and the United States.

5.2. Dollar Regimes and Threshold Estimation

We estimate the dollar regime threshold $\mathcal{D}_\tau$ using the methodology of Hansen (2000). The threshold splits the sample into two regimes based on the level of the broad dollar index D_t .

For each maturity n , we estimate:

$$\Delta x_{n,t} = \alpha_n + \beta_n^L \Delta D_t \mathbf{1}(D_t < \mathcal{D}_\tau) + \beta_n^H \Delta D_t \mathbf{1}(D_t \geq \mathcal{D}_\tau) + \Gamma_n' Z_t + \varepsilon_{n,t}$$

The threshold $\hat{\mathcal{D}}_\tau$ is chosen to minimize the residual sum of squares. Inference follows the bootstrap procedure in Hansen (2000).

5.3. Baseline Threshold Regressions

This specification directly tests Proposition 1 of the model. The theory predicts:

$$\beta_n^L < 0, \quad \beta_n^H \geq \beta_n^L, \quad \text{and often } \beta_n^H \geq 0$$

That is, dollar appreciation widens the basis primarily in low-dollar regimes, while the effect attenuates or reverses when the dollar is already strong.

5.4. Regime-Conditional Local Projections

To examine dynamic effects, we estimate regime-dependent impulse responses using local projections:

$$x_{n,t+h} - x_{n,t-1} = \alpha_n(h) + \beta_n^L(h) \Delta D_t \mathbf{1}(D_t < \mathcal{D}_\tau) + \beta_n^H(h) \Delta D_t \mathbf{1}(D_t \geq \mathcal{D}_\tau) + \Gamma_n(h)' Z_t + \varepsilon_{n,t+h}$$

These responses trace how the basis evolves over time following a dollar shock in different regimes.

5.5. Mechanism Tests: Intermediary Constraints and Tenor Patterns

To link the empirical findings to the theoretical mechanism, we examine whether regime asymmetry is strongest at maturities where Japanese institutional hedging activity is concentrated (5–10 years). We also interact dollar shocks with proxies for intermediary balance sheet constraints to test amplification effects predicted by the model.

6. From Theory to Empirics: Testable Implications for Japan

The theoretical framework developed in Section ?? delivers a sharp empirical prediction: the response of cross-currency basis spreads to dollar movements is ****regime dependent****. The same dollar appreciation can either widen or tighten the basis depending on whether the prevailing environment is characterized by a weak or strong dollar. This nonlinearity arises from regime-dependent shifts in asset-side currency hedging behavior.

The mechanism operates through the hedging decisions of large institutional investors. In low-dollar regimes, investors expect future dollar depreciation and therefore maintain or increase hedge ratios on dollar-denominated assets. When the dollar appreciates in this environment, hedging demand rises, increasing pressure in FX swap markets and widening the basis. In contrast, in high-dollar regimes, expectations shift toward continued dollar strength. Investors are then more willing to remain unhedged, reducing swap-based dollar demand and dampening, or even reversing, the conventional widening effect. The model therefore predicts a **sign reversal** in the dollar–basis relationship across regimes.

Japan provides an especially informative testing ground for this mechanism. Japanese life insurers, pension funds, and banks are among the largest global holders of foreign fixed-income securities, particularly U.S. Treasuries and corporate bonds. These institutions actively manage currency risk using FX swaps and cross-currency basis markets, especially at intermediate and long maturities. As a result, fluctuations in Japanese hedging demand represent a major structural force in global dollar funding markets.

If the theory is correct, three empirical patterns should emerge in the JPY–USD basis:

1. **Regime-dependent sign reversal:** Dollar appreciations should widen the JPY basis in low-dollar regimes but tighten or have muted effects in high-dollar regimes.
2. **Maturity concentration:** The regime asymmetry is strong at across, where Japanese institutional investors are active.
3. **Dynamic persistence:** In low-dollar regimes, basis responses should unfold gradually as hedge ratios adjust over time, while in high-dollar regimes responses should be smaller and less persistent.

To test these predictions, we estimate threshold regressions and regime-conditional local projections for the JPY–USD cross-currency basis across maturities. Regimes are defined using the contemporaneous level of the broad dollar index, consistent with the model’s interpretation of dollar states as prevailing macro-financial environments rather than lagged predictors. The following section presents the results.

7. Main Empirical Results: Dollar Shocks and the JPY Basis

We begin by presenting baseline evidence on how the JPY–USD cross-currency basis responds to dollar movements across regimes. We first estimate a parsimonious specification without additional controls to isolate the raw relationship implied by the model. We then show that the results remain robust after controlling for global risk conditions and relative yield dynamics.

The central finding is a pronounced regime dependence: the sign and magnitude of the dollar–basis relationship differ sharply depending on whether the dollar is in a low or high regime.

7.1. Baseline Threshold Regressions (No Controls)

Table 4 reports discrete threshold regressions across maturities. Dollar regimes are determined endogenously using Hansen (2000) threshold estimation with the broad dollar index (USTWBGD) as the threshold variable.

Across all maturities, dollar appreciations are associated with statistically significant *basis widening* (a more negative basis) when the dollar is in the low regime. In contrast, the

Table 4: Threshold Regressions for Changes in JPY Cross-Currency Basis (No Controls)

	Dependent Variable: Δ JPY Basis				
	(1Y)	(5Y)	(10Y)	(20Y)	(30Y)
<i>Low Dollar Regime</i>					
Threshold (USTWBGD)	<112.87	<111.77	<111.78	<111.77	<111.78
Observations	679	639	640	639	640
Δ USD Index	-1.316*** (0.183)	-0.873*** (0.144)	-0.670*** (0.144)	-0.458*** (0.140)	-0.454*** (0.135)
Constant	-0.074 (0.141)	-0.088 (0.112)	-0.072 (0.112)	-0.038 (0.108)	-0.028 (0.105)
<i>High Dollar Regime</i>					
Threshold (USTWBGD)	$\geq$ 112.87	$\geq$ 111.77	$\geq$ 111.78	$\geq$ 111.77	$\geq$ 111.78
Observations	207	247	246	247	246
Δ USD Index	0.464 (0.291)	0.566*** (0.211)	0.771*** (0.212)	0.633*** (0.204)	0.535*** (0.198)
Constant	0.157 (0.256)	0.096 (0.180)	0.042 (0.181)	-0.055 (0.174)	-0.061 (0.169)
Observations	886	886	886	886	886
R-squared	0.059	0.048	0.038	0.023	0.021
Durbin-Watson	2.21	2.08	2.08	2.16	2.04

Notes: Threshold regressions with regimes determined endogenously using the broad dollar index (USTWBGD). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

relationship weakens or reverses once the dollar enters the high regime, with several maturities showing basis tightening. These results provide direct evidence of the nonlinear, regime-dependent behavior predicted by the model.

7.2. Robustness: Controlling for Risk and Rate Factors

We next assess whether the regime-dependent dollar–basis relationship survives once standard controls are introduced. All regressions include weekly changes in global risk sentiment (VIX), JPY/USD implied volatility, JPY/USD risk reversals, and U.S.–Japan yield and term spread differentials.

Threshold estimation again identifies a stable regime boundary for the broad dollar index at approximately

$$\hat{D}_{JPY}^{\tau} \approx 112,$$

with maturity-specific thresholds tightly clustered around this level. Bootstrap tests reject linearity, confirming statistically significant regime dependence even after including controls.

Table 5: Estimated Dollar Thresholds for JPY Basis Regressions

Maturity	1Y	5Y	10Y	20Y	30Y
Estimated Threshold $\hat{D}^{\tau}$	112.64	111.77	111.78	111.77	111.78
Low-Regime Observations	679	639	640	639	640
High-Regime Observations	207	247	246	247	246
Bootstrap p -value (Linearity Test)	0.05	0.05	0.05	0.05	0.05

Notes: Thresholds are estimated using Hansen (2000) methodology with the contemporaneous broad dollar index as the threshold variable. The table reports maturity-specific threshold estimates, the number of observations classified into low- and high-dollar regimes, and bootstrap p -values for the null hypothesis of linearity. Rejection of linearity supports the presence of regime-dependent dollar effects in JPY basis dynamics.

We estimate

$$\Delta x_t^{JPY}(n) = \beta_L(n) \Delta D_t \mathbf{1}(D_t < \hat{D}^{\tau}) + \beta_H(n) \Delta D_t \mathbf{1}(D_t \geq \hat{D}^{\tau}) + \delta' Z_t + u_t,$$

where Z_t contains the full control set.

Table 6: JPY–USD Basis: Regime-Dependent Dollar Effects with Controls

	1Y	5Y	10Y	20Y	30Y
ΔD_t (Low Regime)	-0.70***	-0.72***	-0.66***	-0.50***	-0.49***
ΔD_t (High Regime)	0.87***	0.66***	0.77***	0.60***	0.50***
ΔVIX_t	-0.25***	-0.04	0.02	0.01	0.02
ΔRR_t	0.78**	0.26	0.47	0.47	0.27
ΔIV_t	0.38	-0.19	0.03	0.20	0.06
$\Delta YSPRD_t$	2.03	2.11	1.57	0.85	1.14
$\Delta TSPRD_t$	0.15	-1.72	-1.82	-1.69	-0.95
Adj. R^2	0.14	0.05	0.03	0.02	0.01

The regime-dependent sign reversal remains economically large. Regime asymmetry is strong across all maturities. At the 10-year maturity, the dollar slope shifts from -0.66 in

low-dollar states to +0.77 in high-dollar states. The same striking reversal is true at all maturities. The control variables behave in economically intuitive ways. Increases in global risk aversion (VIX) widen the short-end basis, while FX volatility measures play a limited role once regimes are accounted for. Yield spread and term spread changes do not compress the basis in statistically significant ways across all tenors. Importantly, inclusion of these controls does not alter the regime-dependent dollar coefficients, confirming that the nonlinear dollar–basis relationship reflects hedging behavior rather than omitted risk or rate factors. Crucially, the inclusion of these controls leaves the regime-dependent dollar coefficients largely unchanged, indicating that the nonlinear dollar–basis relationship reflects hedging behavior rather than omitted macro-financial factors.

7.3. Dynamic Responses: Regime-Conditional Local Projections

While the threshold regressions establish that the contemporaneous response of the JPY–USD basis to dollar shocks differs sharply across regimes, they do not reveal how these effects evolve over time. If the mechanism operates through gradual adjustments in institutional FX hedge ratios, the basis response should be persistent in low-dollar states but muted in high-dollar states. To examine these dynamics, we estimate regime-conditional local projections.

For each maturity n and forecast horizon $h = 0, 1, \dots, 8$ weeks, we estimate

$$x_{t+h}^{JPY} - x_{t-1}^{JPY} = \beta_L^{(h)} \Delta D_t \mathbf{1}(D_t < \hat{D}^\tau) + \beta_H^{(h)} \Delta D_t \mathbf{1}(D_t \geq \hat{D}^\tau) + \Gamma_h' Z_t + \varepsilon_{t+h},$$

where x_t^{JPY} denotes the JPY–USD cross-currency basis, ΔD_t is the weekly change in the broad dollar index, and Z_t contains the full set of controls (changes in VIX, JPY/USD implied volatility, risk reversals, and U.S.–Japan yield and term spread differentials). Standard errors are Newey–West adjusted to account for serial correlation induced by overlapping horizons.

Figure 3 plots the impulse responses $\beta_L^{(h)}$ and $\beta_H^{(h)}$ across horizons for representative maturities, with 95% confidence bands.

Low-dollar regime dynamics.. In low-dollar states, dollar appreciations generate a persistent widening of the JPY basis (a move toward a more negative basis). The response builds gradually over time, typically reaching its peak between two and four weeks after the initial shock before slowly dissipating. This delayed amplification is consistent with gradual hedge-ratio adjustments by Japanese institutional investors. Life insurers and pension funds typically rebalance currency hedges with institutional frictions, internal risk limits, and portfolio management lags, implying that hedge adjustments unfold over time rather than instantaneously.

High-dollar regime dynamics.. In contrast, in high-dollar states the basis response is substantially weaker and short-lived. Dollar appreciations initially produce modest basis tightening, but the effect fades quickly and becomes statistically indistinguishable from zero after only a few weeks. The lack of persistence suggests limited hedge-ratio adjustment when the dollar is already strong and investors are more willing to tolerate unhedged foreign-currency exposures.

Regime contrast.. The divergence between regimes is economically large. Over a one-month horizon, the cumulative response in the low-dollar regime remains strongly negative (basis widening), while the cumulative response in the high-dollar regime is close to zero or slightly positive. This sharp difference in persistence provides dynamic evidence that regime-dependent institutional hedging behavior, rather than contemporaneous funding stress alone, drives the nonlinear behavior of the JPY basis.

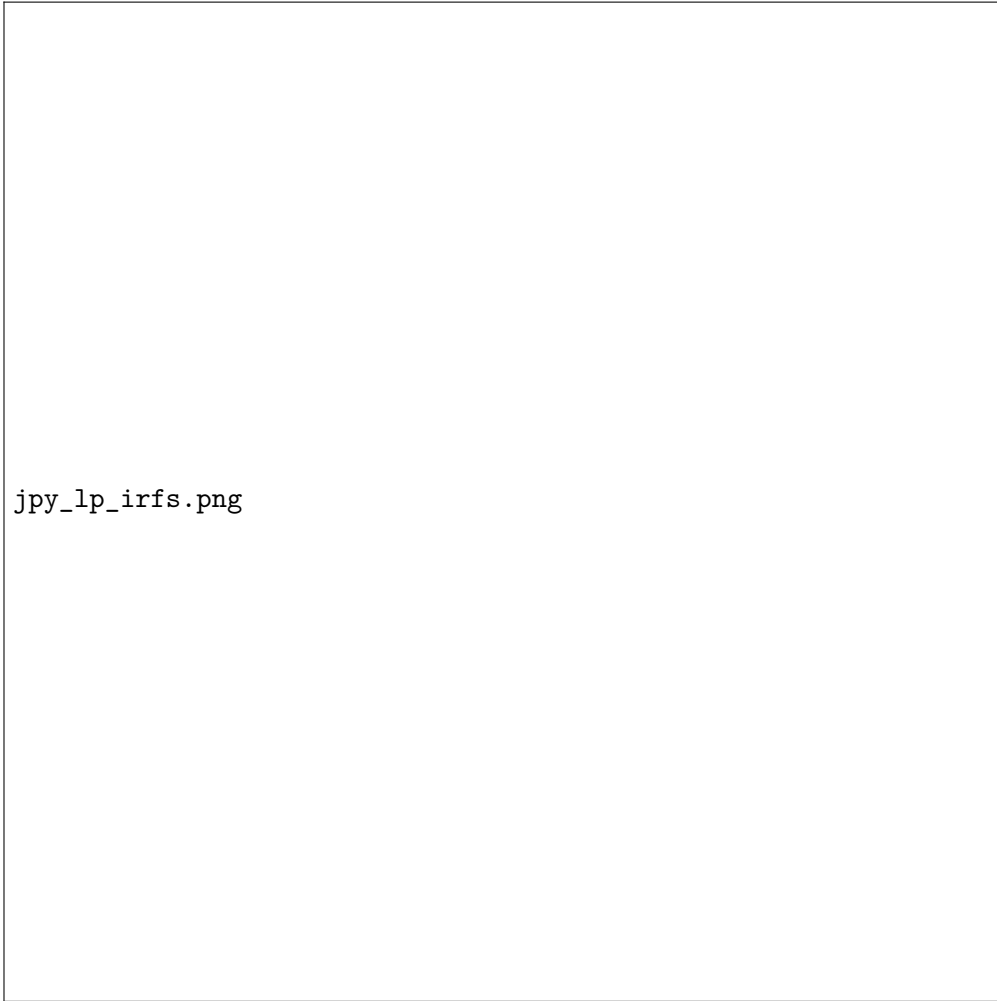


Figure 3: Regime-Conditional Local Projection Responses of the JPY Basis to Dollar Shocks
Notes: Impulse responses of the JPY–USD cross-currency basis to a one-standard-deviation dollar appreciation shock. Solid lines show point estimates; shaded areas denote 95% confidence intervals. Blue lines correspond to the low-dollar regime ($D_t < \hat{D}^\tau$), and red lines correspond to the high-dollar regime ($D_t \geq \hat{D}^\tau$).

8. Mechanism Evidence: Hedging Demand, Portfolio Flows, and Intermediary Amplification

This section provides direct evidence for the model’s mechanism: regime dependence in the dollar–basis relationship reflects *state-dependent net hedging pressure* rather than funding stress alone. Proposition 2 links the dollar–basis slope to regime-specific hedging intensity,

$$\beta_t^\theta \equiv \frac{\partial x_t}{\partial u_t} \Big|_{\theta_t=\theta} = -\frac{\Lambda}{1 + \Lambda \Psi_t} \Xi_t^\theta, \quad \Xi_t^\theta \equiv A_t^F \eta^\theta - A_t^U \xi^\theta, \quad (20)$$

so that a dollar shock moves the basis through the difference between (i) *USD-demanding hedgers* (Japanese investors hedging USD assets) and (ii) *USD-supplying hedgers* (USD investors hedging foreign assets). We test three complementary implications: (i) measures that increase the incentive to *hedge USD assets* should predict basis widening especially in the low-dollar regime; (ii) increases in USD asset inflows should predict basis widening through the same channel; and (iii) intermediary tightness should amplify (but not overturn) the regime-specific mapping from dollar shocks into the basis.

8.1. Hedged Yield Pickup and Hedging Incentives

8.1.1. Measure

A natural proxy for the incentive to hedge USD bonds is the *hedged yield pickup* for an investor based in currency i who purchases a USD bond and hedges FX risk back into i :

$$\text{HYP}_{i,t} \equiv y_t^\$ - (y_{i,t} - x_{i,t}), \quad (21)$$

where $y_t^\$$ is the USD yield at maturity n , $y_{i,t}$ is the local-currency yield at the same maturity, and $x_{i,t}$ is the cross-currency basis (with $x_{i,t} < 0$ indicating that USD is expensive in swaps). Intuitively, $(y_{i,t} - x_{i,t})$ is the *hedged* yield available to an investor funding in currency i and swapping into USD, so $\text{HYP}_{i,t}$ is the return advantage of holding USD bonds on a currency-hedged basis.

¹

¹To see why this expression captures the incentive to hedge USD bonds, consider an investor based in currency i . The investor has two choices: (i) invest domestically and earn the local yield $y_{i,t}$, or (ii) buy a USD bond yielding $y_t^\$$ and fully hedge the exchange-rate risk back into currency i using FX swaps. When covered interest parity does not hold, the FX swap embeds a cross-currency basis $x_{i,t}$. Under the convention used here, $x_{i,t} < 0$ means USD is expensive to obtain via swaps, so hedging USD exposure is costly. The hedged return on the USD bond, expressed in currency i , is therefore $y_t^\$ + x_{i,t}$. The return advantage of the hedged USD bond relative to staying domestic is

$$(y_t^\$ + x_{i,t}) - y_{i,t} = y_t^\$ - (y_{i,t} - x_{i,t}),$$

which is exactly $\text{HYP}_{i,t}$. A higher $\text{HYP}_{i,t}$ means that USD bonds offer a larger return pickup on a currency-hedged basis, increasing the incentive for foreign investors to hold and hedge USD assets. This, in turn, raises demand for USD in FX swap markets and puts downward (more negative) pressure on the basis. In the model, this incentive operates through the hedge ratio of foreign investors, h_t^F , and therefore through total USD hedging demand $Q_t = A_t^F h_t^F$. When $\text{HYP}_{i,t}$ rises, the optimal hedge component of h_t^F increases, raising Q_t , increasing net USD demand in FX swaps, and putting downward (more negative) pressure on the cross-currency basis.

8.1.2. Identification issue and orthogonalized pickup

Because $\text{HYP}_{i,t}$ mechanically embeds the basis (it increases one-for-one with $x_{i,t}$), levels of $\text{HYP}_{i,t}$ cannot be used to explain levels of $x_{i,t}$ without inducing a tautology. To isolate *incentives* rather than mechanical accounting, we adopt two complementary approaches.

First, we focus on *changes* and exclude the contemporaneous basis component:

$$\widetilde{\Delta\text{HYP}}_{i,t} \equiv \Delta(y_t^{\$} - y_{i,t}), \quad (22)$$

so that $\widetilde{\Delta\text{HYP}}_{i,t}$ captures movements in relative yields that plausibly shift hedging incentives but does not mechanically include $\Delta x_{i,t}$.

Second, as a robustness check, we form an *orthogonalized* pickup by partialling out the basis component:

$$\Delta\text{HYP}_{i,t}^{\perp} = \Delta\text{HYP}_{i,t} - \hat{\rho} \Delta x_{i,t}, \quad (23)$$

where $\hat{\rho}$ is obtained from a first-stage regression of $\Delta\text{HYP}_{i,t}$ on $\Delta x_{i,t}$ (estimated within maturity and currency). Both constructions isolate changes in yield differentials that move incentives to hedge USD assets without mechanically importing the dependent variable.

8.1.3. Hypothesis

If regime dependence reflects asset-side hedging, then increases in hedging incentives—as proxied by $\widetilde{\Delta\text{HYP}}_{i,t}$ or $\Delta\text{HYP}_{i,t}^{\perp}$ —should coincide with *greater USD-demanding hedging*, i.e., higher $A_t^F \eta^{\theta}$ in (20). The model implies this link should be strongest in the low-dollar regime, where Japanese institutions mechanically rebalance hedge ratios more aggressively following dollar appreciations.

8.1.4. Empirical specification

We estimate regime-specific panel regressions of the form

$$\Delta x_{i,t} = \sum_{\theta \in \{L, H\}} \left(\alpha_{\theta} + \beta_{\theta} \Delta D_t + \phi_{\theta} \Delta \mathcal{H}_{i,t} + \psi_{\theta} \Delta D_t \cdot \Delta \mathcal{H}_{i,t} + \Gamma'_{\theta} Z_t \right) \mathbf{1}(\theta_t = \theta) + \varepsilon_{i,t}, \quad (24)$$

where $\Delta \mathcal{H}_{i,t}$ denotes either $\widetilde{\Delta\text{HYP}}_{i,t}$ or $\Delta\text{HYP}_{i,t}^{\perp}$, D_t is the broad dollar, and Z_t includes the baseline controls used in Section 5. Standard errors are heteroskedasticity-robust and clustered by time (and, in panel specifications, additionally by currency where applicable).

8.1.5. Predictions

The hedging-demand mechanism yields three sharp predictions.

1. **Low-dollar sensitivity:** $\phi_L < 0$. When hedging incentives rise, the basis widens (more negative $x_{i,t}$) in low-dollar states.
2. **Attenuation/reversal in high-dollar states:** $|\phi_L| > |\phi_H|$ and potentially $\phi_H \geq 0$, consistent with a weaker hedge-ratio response when the dollar is already strong.
3. **Amplification of dollar shocks:** $\psi_L < 0$. Dollar appreciations widen the basis more when hedging incentives are rising, isolating a direct “hedging-demand amplification” signature.

8.1.6. Interpretation

A negative ϕ_L and ψ_L are difficult to rationalize with funding stress alone. Funding-based stories predict that the basis widens when intermediaries are constrained, but do not naturally imply that *increases in hedging incentives* (holding funding conditions fixed) predict incremental basis widening *specifically* in low-dollar states. By contrast, (20) implies that any variable that increases the intensity of USD-demanding hedgers (higher $A_t^F \eta^L$) should widen the basis precisely in those states.

8.2. Portfolio Flow Proxies: USD Asset Inflows and Basis Widening

8.2.1. Motivation

If Japanese demand for USD assets rises—especially in Treasuries and long-duration USD bonds—then associated FX hedging demand should increase, raising net USD demand in swaps and widening the basis. In the model, this corresponds to a rise in exposure A_t^F and/or a higher marginal rebalancing intensity η^θ , both of which raise Ξ_t^θ in (20) and therefore steepen the dollar–basis slope in low-dollar regimes.

8.2.2. Flow measures and scaling

Let Flow_t denote a proxy for net purchases of USD fixed-income assets by Japanese investors (or, in broader specifications, foreign investors), constructed from data such as Treasury International Capital (TIC) transactions, Balance of Payments portfolio flows, or custody/reserve-manager holdings where available. Because the mechanism concerns *changes in net hedging pressure*, we focus on *changes in flows* (or net flows scaled by a size measure such as lagged holdings, AUM, or GDP):

$$\Delta \text{Flow}_t, \quad \text{or} \quad \Delta \text{Flow}_t^s \equiv \frac{\text{Flow}_t}{\text{Size}_{t-1}}. \quad (25)$$

This choice aligns the empirical proxy with the model object ΔA_t^F and avoids spurious correlation from trending holdings or persistent level effects.

8.2.3. Empirical specification

We estimate

$$\Delta x_t = \sum_{\theta \in \{L, H\}} \left(\alpha_\theta + \beta_\theta \Delta D_t + \varphi_\theta \Delta \text{Flow}_t + \psi_\theta \Delta D_t \cdot \Delta \text{Flow}_t + \Gamma'_\theta Z_t \right) \mathbf{1}(\theta_t = \theta) + \varepsilon_t, \quad (26)$$

where for the JPY–USD focus Δx_t is the change in the JPY basis at maturity n . When flow measures are available at lower frequency (e.g., monthly), we align the basis and dollar variables to the same frequency and include appropriate lag structures.

8.2.4. Predictions

The net-hedging mechanism implies:

1. **Inflow–widening link in low-dollar states:** $\varphi_L < 0$.
2. **Interaction signature:** $\psi_L < 0$. Dollar appreciation combined with increasing USD inflows produces additional basis widening.
3. **Attenuation in high-dollar states:** φ_H and ψ_H attenuate toward zero and may change sign if the net hedging configuration shifts (lower η^H and/or higher ξ^H).

8.2.5. Interpretation

The interaction term ψ_θ is the key mechanism test. It isolates whether dollar shocks have larger price impact precisely when USD asset inflows are rising—the empirical counterpart of (20), where dollar shocks transmit through Ξ_t^θ and are amplified when the exposure and/or hedge-ratio response of USD-demanding hedgers is increasing.

8.3. Intermediary Constraints as an Amplification Layer

8.3.1. Motivation

Proposition 3 implies that the dollar-basis slope is scaled by intermediary tightness:

$$|\beta_t^\theta| = \frac{\Lambda}{1 + \Lambda\Psi_t} |\Xi_t^\theta|, \quad \frac{\partial}{\partial \Lambda} |\beta_t^\theta| > 0. \quad (27)$$

Thus tighter intermediary constraints should amplify the regime-specific basis response to dollar shocks, without changing the sign logic pinned down by Ξ_t^θ .

8.3.2. Proxies

We proxy intermediary tightness, Λ_t , using measures designed to capture swap-market and dealer balance-sheet conditions, such as: (i) money-market stress spreads (e.g., SOFR–IORB or related spreads), (ii) primary dealer leverage/balance-sheet indicators, (iii) Treasury market depth or illiquidity measures, (iv) intermediary equity returns (e.g., broker-dealer indices), and (v) swap-spread dislocations that are known to be balance-sheet sensitive. Each proxy is standardized so that higher values indicate tighter constraints.

8.3.3. Empirical specification

We estimate

$$\Delta x_t = \sum_{\theta \in \{L, H\}} \left(\alpha_\theta + \beta_\theta \Delta D_t + \beta_{\Lambda, \theta} (\Delta D_t \cdot \Lambda_t) + \Gamma'_\theta Z_t \right) \mathbf{1}(\theta_t = \theta) + \varepsilon_t. \quad (28)$$

8.3.4. Predictions and interpretation

The amplification layer implies:

$$\text{sign}(\beta_{\Lambda, \theta}) = \text{sign}(\beta_\theta), \quad \text{and} \quad |\beta_\theta + \beta_{\Lambda, \theta} \Lambda_t| \text{ increases in } \Lambda_t. \quad (29)$$

This test does not “replace” hedging demand. Instead, it verifies the model’s decomposition: hedging imbalance determines the *direction* of the basis response through Ξ_t^θ , while intermediary constraints scale the *magnitude* through Λ_t .

8.4. Discussion: Distinguishing Hedging from Funding Stress

Taken together, the three tests map directly into the theory. The hedged yield pickup proxy targets movements in hedging *incentives*; portfolio flows target changes in hedging *exposures*; and intermediary proxies target the pricing of net order flow through balance-sheet constraints. A pure funding-stress account can explain basis widening in crisis episodes, but it does not naturally predict that (i) changes in hedging incentives and (ii) increases in USD asset inflows should *selectively* strengthen basis widening in low-dollar regimes via the interaction terms in (24) and (26). By contrast, the model predicts exactly these regime-specific amplification signatures because they increase Ξ_t^θ when η^L is high and hedge-ratio rebalancing is most active.

8.5. Summary of Findings

For Japan, the JPY–USD basis exhibits a clear regime-dependent sign reversal in response to dollar shocks. Low-dollar regimes produce persistent basis widening; high-dollar regimes produce tightening or negligible effects. The pattern aligns with regime-dependent institutional hedging, with intermediary constraints acting as an amplification layer.

The sign reversal remains economically large. At the 10-year maturity, the slope shifts from -0.66 in low-dollar states to $+0.77$ in high-dollar states, confirming that the nonlinear response is not driven by omitted risk or rate factors.

8.6. Dynamic Responses: Regime-Conditional Local Projections

We next examine the dynamic adjustment of the JPY basis following a dollar appreciation using regime-conditional local projections.

Low-dollar regime dynamics.. In low-dollar states, dollar appreciations generate a *persistent widening* of the JPY basis. The impulse responses show that the basis continues to move in the widening direction for several weeks following the initial shock, with peak effects typically occurring 2–4 weeks after impact. This gradual adjustment is consistent with the model’s interpretation that Japanese institutional investors increase FX hedge ratios over time in response to dollar movements, rather than adjusting instantaneously.

High-dollar regime dynamics.. In contrast, in high-dollar states, the basis response is small and often reverses within a few weeks. Dollar appreciations initially lead to modest basis tightening, but the effect dissipates quickly. The lack of persistence suggests that hedge-ratio adjustments by Japanese investors are muted when the dollar is already strong, consistent with greater tolerance for unhedged foreign exposures in such environments.

Regime contrast.. The difference in cumulative impulse responses across regimes is substantial. Over a one-month horizon, the cumulative low-regime response remains strongly negative (basis widening), while the cumulative high-regime response is close to zero or positive. This divergence directly supports the model’s mechanism of regime-dependent net hedging intensity.

8.7. Summary of Findings

The empirical evidence for Japan reveals a clear and economically large regime-dependent sign reversal in the response of the JPY–USD basis to dollar shocks. In low-dollar regimes, appreciations of the dollar generate persistent basis widening; in high-dollar regimes, they generate basis tightening or negligible effects. The asymmetry is strongest at maturities associated with Japanese institutional hedging and is amplified when intermediary constraints tighten.

These findings strongly support the model’s central implication: nonlinear basis dynamics arise from regime-dependent asset-side hedging behavior, with intermediary balance sheet constraints acting as an amplification layer rather than the primary source of nonlinearity.

9. Conclusion

This paper shows that the relationship between U.S. dollar movements and deviations from covered interest parity is not stable even within a single, economically central market: the JPY–USD cross-currency basis. Using weekly data across maturities from 1 to 30 years, we document a pronounced state dependence in the response of the JPY basis to dollar shocks. When the dollar is in a low regime, dollar appreciations produce economically large and statistically significant basis widening (a more negative basis). When the dollar is already high, the same dollar appreciation generates a much smaller response and frequently leads to basis tightening. This regime dependence is not a minor nuance: across maturities, the estimated dollar slope switches sign between low- and high-dollar environments, and the asymmetry is strongest in the 5–10 year sector, where Japanese life insurers and pension funds concentrate their hedged U.S. bond exposures.

We interpret these patterns through a simple equilibrium framework in which the basis is the price that clears net currency-hedging demand in the presence of intermediary balance sheet costs. The key ingredient is regime-dependent asset-side hedging by Japanese institutional investors. When the dollar is moderate or weak, Japanese institutions tend to maintain higher hedge ratios on their large USD bond portfolios. In that environment, a dollar appreciation triggers additional hedging demand—through risk management rules, solvency constraints, and asset–liability management—raising net demand for dollars in swap markets and widening the basis. When the dollar is already strong, expectations and hedging incentives shift: investors are more willing to remain partially unhedged or to accept dollar carry, so hedging demand responds less, and net order flow can even move in the opposite direction, producing basis tightening. Intermediary constraints remain important, but in our framework they operate primarily as an amplification layer: tighter dealer balance sheets increase the mapping from net hedging pressure into the equilibrium basis without determining the sign of the response.

Our empirical results support this interpretation along three dimensions. First, threshold regressions based on Hansen (2000) identify a remarkably stable dollar regime boundary around a broad-dollar level near 112, and the estimated dollar slopes exhibit a sharp sign reversal across regimes. Second, these findings are robust to controlling for global risk sentiment (VIX), FX option-implied volatility and skew (implied volatility and risk reversals), and U.S.–Japan yield and term spread differentials. The inclusion of these controls leaves the regime-dependent dollar coefficients largely unchanged, indicating that the nonlinearity is not an artifact of omitted risk conditions or rate differentials. Third, regime-conditional local projections reveal that the dynamics differ across regimes in a way that matches the hedging mechanism: in low-dollar states the basis response is persistent and peaks with a delay of roughly two to four weeks, consistent with gradual hedge-ratio adjustment, whereas in high-dollar states the response is smaller and short-lived.

These findings have two broader implications. The first is conceptual: the JPY–USD cross-currency basis should be interpreted not only as a measure of dollar funding stress, but also as an equilibrium price of institutional currency hedging. The same dollar appreciation can correspond to very different underlying drivers depending on the prevailing dollar regime. The second is practical: regime dependence changes the informational content of basis movements for policymakers and market participants. In low-dollar environments, basis widening following dollar appreciation is more likely to reflect intensified hedging demand and the associated pressure on swap-based dollar intermediation. In high-dollar environments, muted

or reversed basis responses suggest that institutional hedging incentives have shifted, so basis movements are a less reliable real-time proxy for marginal funding stress.

Several extensions follow naturally. A first priority is to connect the regime mechanism more directly to institution-level hedging activity using portfolio flow measures, hedged foreign bond returns, or supervisory data on hedge ratios for Japanese life insurers and pensions. A second extension is to jointly model hedging demand and dealer constraints in a multi-country setting to quantify how shocks to one large hedger group propagate across other currency bases. Finally, incorporating forward-looking expectations and endogenous regime transitions would sharpen the mapping between dollar states, hedging incentives, and basis dynamics. Overall, the evidence presented here highlights that nonlinear FX pricing in swap markets is tightly linked to the state of the dollar cycle and to the portfolio rebalancing behavior of large institutional investors, with Japan playing a central role in the international transmission of dollar shocks.

Appendix A: Proofs

This appendix provides formal proofs for Propositions 1–3. We begin by deriving the optimal hedge ratios, then solve for equilibrium net order flow and the cross-currency basis.

A.1. Mean–variance hedge demands

Foreign investor (USD asset holder)

A foreign investor holds a USD asset and chooses hedge ratio h_t^F . Foreign-currency excess return from the position is approximated by

$$\tilde{R}_{t+1}^F(h) = \bar{R}_t^F - (1-h)\Delta s_{t+1} - h\pi_t. \quad (.1)$$

Expected return and variance are

$$\mathbb{E}_t[\tilde{R}_{t+1}^F(h)] = \bar{R}_t^F - (1-h)\mu_t^\theta - h\pi_t, \quad \text{Var}_t(\tilde{R}_{t+1}^F(h)) = (1-h)^2\sigma_{s,t}^2.$$

The investor solves

$$\max_h \mathbb{E}_t[\tilde{R}_{t+1}^F(h)] - \frac{\gamma_F}{2} \text{Var}_t(\tilde{R}_{t+1}^F(h)).$$

FOC:

$$\mu_t^\theta - \pi_t + \gamma_F(1-h)\sigma_{s,t}^2 = 0 \quad \Rightarrow \quad h_t^{F,opt} = 1 + \frac{\mu_t^\theta - \pi_t}{\gamma_F\sigma_{s,t}^2}.$$

USD investor (foreign asset holder)

A USD investor holds a foreign asset. USD return approximation:

$$\tilde{R}_{t+1}^U(h) = \bar{R}_t^U + (1-h)\Delta s_{t+1} - h\pi_t^U. \quad (.2)$$

Expected return and variance:

$$\mathbb{E}_t[\tilde{R}_{t+1}^U(h)] = \bar{R}_t^U + (1-h)\mu_t^\theta - h\pi_t^U, \quad \text{Var}_t(\tilde{R}_{t+1}^U(h)) = (1-h)^2\sigma_{s,t}^2.$$

FOC gives

$$-\mu_t^\theta - \pi_t^U + \gamma_U(1-h)\sigma_{s,t}^2 = 0 \quad \Rightarrow \quad h_t^{U,opt} = 1 - \frac{\mu_t^\theta + \pi_t^U}{\gamma_U\sigma_{s,t}^2}.$$

A.2. Net USD order flow

Total USD demand and supply in swap markets:

$$Q_t = A_t^F h_t^F, \quad R_t = A_t^U h_t^U,$$

so net order flow

$$N_t = Q_t - R_t.$$

Substitute hedge rules with regime-dependent mechanical components:

$$h_t^F = 1 + \frac{\mu_t^\theta - \pi_t}{\gamma_F\sigma_{s,t}^2} + \eta^\theta u_t,$$

$$h_t^U = 1 - \frac{\mu_t^\theta + \pi_t^U}{\gamma_U\sigma_{s,t}^2} + \xi^\theta u_t.$$

Using $\pi_t = \pi_0 - \kappa x_t$ and $\pi_t^U = \pi_0^U + \kappa_U x_t$,

$$N_t = A_t^F - A_t^U \tag{.3}$$

$$+ \left(\frac{A_t^F}{\gamma_F\sigma_{s,t}^2} + \frac{A_t^U}{\gamma_U\sigma_{s,t}^2} \right) \mu_t^\theta \tag{.4}$$

$$+ \left(\frac{-A_t^F\pi_0}{\gamma_F\sigma_{s,t}^2} + \frac{A_t^U\pi_0^U}{\gamma_U\sigma_{s,t}^2} \right) \tag{.5}$$

$$+ \left(\frac{A_t^F\kappa}{\gamma_F\sigma_{s,t}^2} + \frac{A_t^U\kappa_U}{\gamma_U\sigma_{s,t}^2} \right) x_t \tag{.6}$$

$$+ (A_t^F\eta^\theta - A_t^U\xi^\theta) u_t. \tag{.7}$$

Define

$$\Psi_t = \frac{A_t^F\kappa}{\gamma_F\sigma_{s,t}^2} + \frac{A_t^U\kappa_U}{\gamma_U\sigma_{s,t}^2}, \quad \Xi_t^\theta = A_t^F\eta^\theta - A_t^U\xi^\theta,$$

$$\Omega_t^\theta = (A_t^F - A_t^U) + \left(\frac{A_t^F}{\gamma_F\sigma_{s,t}^2} + \frac{A_t^U}{\gamma_U\sigma_{s,t}^2} \right) \mu_t^\theta + \left(\frac{-A_t^F\pi_0}{\gamma_F\sigma_{s,t}^2} + \frac{A_t^U\pi_0^U}{\gamma_U\sigma_{s,t}^2} \right).$$

Then

$$N_t = \Omega_t^\theta + \Psi_t x_t + \Xi_t^\theta u_t.$$

A.3. Equilibrium basis

Dealer clearing:

$$x_t = -\Lambda N_t.$$

2

Substitute:

$$N_t = \Omega_t^\theta - \Lambda \Psi_t N_t + \Xi_t^\theta u_t,$$

so

$$(1 + \Lambda \Psi_t) N_t = \Omega_t^\theta + \Xi_t^\theta u_t.$$

Thus

$$N_t = \frac{\Omega_t^\theta + \Xi_t^\theta u_t}{1 + \Lambda \Psi_t}, \quad x_t = -\frac{\Lambda}{1 + \Lambda \Psi_t} \Omega_t^\theta - \frac{\Lambda}{1 + \Lambda \Psi_t} \Xi_t^\theta u_t.$$

This proves Proposition 1.

A.4. Proof of Proposition 2

Differentiate equilibrium basis with respect to dollar shock:

$$\beta_t^\theta = \frac{\partial x_t}{\partial u_t} = -\frac{\Lambda}{1 + \Lambda \Psi_t} \Xi_t^\theta.$$

Since $\Lambda > 0$, the sign of β_t^θ is opposite the sign of Ξ_t^θ , yielding the sign conditions in Proposition 2.

A.5. Proof of Proposition 3

Let $f(\Lambda) = \Lambda/(1 + \Lambda \Psi_t)$. Then

²**Microfoundation for $x_t = -\Lambda N_t$.** Let N_t denote net USD order flow in the FX swap market (USD demanded by foreign hedgers minus USD supplied by USD hedgers). A dealer (or competitive dealer sector) must absorb this imbalance on balance sheet by taking the opposite swap position. Intermediation uses scarce balance-sheet capacity and is costly because it consumes funding, leverage, and risk limits. A standard reduced-form way to capture this is to assume a convex balance-sheet cost of warehousing flow,

$$\mathcal{C}(N_t) = \frac{\varphi}{2} N_t^2, \quad \varphi > 0,$$

so the marginal intermediation cost is $\mathcal{C}'(N_t) = \varphi N_t$. Competitive or near-competitive pricing implies that the *per-unit* compensation embedded in swap pricing (the basis) is proportional to this marginal cost, up to a pass-through factor $\lambda > 0$ that captures market structure and pricing frictions (e.g., dealer markups, segmentation, and imperfect competition):

$$-x_t = \lambda \mathcal{C}'(N_t) = \lambda \varphi N_t.$$

Define $\Lambda \equiv \lambda \varphi > 0$, yielding

$$x_t = -\Lambda N_t.$$

The negative sign follows the standard convention that $x_t < 0$ corresponds to a wider (more negative) basis: when net USD demand is positive ($N_t > 0$), dealers must supply USD via swaps and therefore require a larger compensation, which appears as a *more negative* basis. Thus φ governs the curvature/tightness of intermediary constraints, λ governs how strongly marginal balance-sheet costs are transmitted into prices, and Λ summarizes the overall slope of the dealer supply schedule in the swap market. Allowing Λ to vary over time is equivalent to allowing intermediary conditions (balance-sheet tightness and/or pricing power) to change.

$$f'(\Lambda) = \frac{1}{(1 + \Lambda\Psi_t)^2} > 0.$$

Thus $|\beta_t^\theta| = f(\Lambda)|\Xi_t^\theta|$ increases in Λ .

□

Appendix B: Additional Empirical Results

[Place your additional tables/figures here.]

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